UNIT - II

DISASTERS

Course Code: 22CE409

DISASTER PREPAREDNESS, PLANNING AND MANAGEMENT

(Open Elective-I: Common to all branches except CE)

Instruction	:	3 Periods / week	Continuous Internal Evaluation	:	40 Marks
Tutorial	:	0 Period / week	Semester End Examination	:	60 Marks
Credits	:	3	Semester End Exam Duration	:	3 Hours

Prerequisites:

Course Objectives: The objectives of the course are

1. To Understand basic concepts in Disaster Management.
2. To Understand Definitions and Terminologies used in Disaster Management.
3. To Understand Types and Categories of Disasters.
4. To Understand the Challenges posed by Disasters.
5. To understand Impacts of Disasters Key Skills.

Unit-I

Introduction- Concepts and definitions: disaster, hazard, vulnerability, resilience, risks severity, frequency and details, capacity, impact, prevention, mitigation, disaster phenomena, events-global National & Regional.

Unit-II

Disasters- Disasters classification; natural disasters (floods, draught, cyclones, volcanoes, earthquakes, tsunami, landslides, coastal erosion, soil erosion, forest fires etc.); manmade disasters(industrial pollution, artificial flooding in urban areas, nuclear radiation, chemical spills, transportation accidents, terrorist strikes, etc.); hazard and vulnerability profile of India, Caved 2019 in India, mountain and coastal areas, ecological fragility, coping with disaster-strategies, safety norms &survival kits.

Unit-III

Disaster Impacts- Disaster impacts (environmental, physical, social, ecological, economic, political, etc.); health, psycho-social issues; demographic aspects (gender, age, special needs); hazard locations; global and national disaster trends; climate change and urban disasters, capacity building – concepts, assessment –structural & non-structural measures, legislative support.

Natural Disasters are catastrophic events caused by natural processes or phenomena that result in significant damage to life, property, and the environment. These disasters are often sudden and can have varying levels of severity depending on the location, preparedness, and resilience of the affected areas.

Floods

Definition:

Floods are a natural disaster characterized by the overflow of water onto normally dry land, submerging areas and causing widespread damage to property, infrastructure, and loss of life. They occur when the volume of water exceeds the capacity of natural or man-made drainage systems, such as rivers, lakes, or reservoirs.

Causes of Floods:

1.Natural Causes:

1. **Heavy Rainfall**: Prolonged or intense rainfall leads to rivers and streams overflowing their banks.
2. **Snowmelt**: Rapid melting of snow due to sudden temperature increases contributes to excess water flow in rivers.
3. **Storm Surges**: Cyclones and hurricanes generate storm surges that inundate coastal areas.
4. **Tsunamis**: Large ocean waves triggered by underwater earthquakes can flood coastal regions.
5. **Topography**: Flat or low-lying areas are naturally more prone to flooding.

2. Man-Made Causes:

•**Dam Failure**: The collapse of dams due to structural weakness or excessive water pressure.

•**Urbanization**: Increased impermeable surfaces (like roads and buildings) reduce natural water absorption, leading to urban floods.

•**Deforestation**: Loss of forest cover reduces water absorption capacity, leading to higher surface runoff.

•**Improper Drainage Systems**: Blocked or inadequate drainage systems in cities can lead to waterlogging.

Types of Floods:

1. **River Floods**:
    1. Caused by the overflow of rivers due to heavy rainfall or snowmelt.
    2. Example: Brahmaputra River floods in Assam, India.
2. **Flash Floods**:
    1. Sudden, short-term floods caused by heavy localized rainfall or dam bursts.
    2. Extremely dangerous due to the speed and force of water flow.
3. **Coastal Floods**:
    1. Occur in coastal areas due to storm surges, tsunamis, or high tides.
    2. Example: Coastal floods during Cyclone Amphan in India.
4. **Urban Floods**:
    1. Caused by inadequate drainage systems in cities during heavy rains.
    2. Example: Floods in Mumbai during monsoons.
5. **Pluvial Floods**:
    1. Caused by heavy rainfall pooling in flat areas without adequate drainage.
    2. Often seen in agricultural lands.
6. **Glacial Lake Outburst Floods (GLOF)**:
    1. Caused by the sudden release of water from a glacial lake due to melting ice or landslides.

Impacts of Floods:

1. **Social Impacts**:
    1. Loss of lives and displacement of communities.
    2. Spread of waterborne diseases like cholera and typhoid.
2. **Economic Impacts**:
    1. Destruction of infrastructure, homes, and industries.
    2. Agricultural losses due to submerged farmlands.
3. **Environmental Impacts**:
    1. Erosion of soil and sediment deposition.
    2. Loss of biodiversity due to destruction of habitats.

Drought

Definition:

Drought is a prolonged period of abnormally low rainfall, leading to water scarcity, reduced soil moisture, and adverse impacts on agriculture, ecosystems, and human activities. It is one of the most devastating natural disasters due to its slow onset and widespread effects.

Causes of Drought:

1. **Natural Causes**:
    1. **Low Rainfall**: Prolonged dry weather or failure of monsoon rains leads to drought conditions.
    2. **Climate Change**: Rising global temperatures increase evaporation and reduce precipitation.
    3. **El Niño Effect**: Disruption of normal weather patterns due to oceanic temperature changes can lead to reduced rainfall.
    4. **Geographical Factors**: Arid and semi-arid regions are more prone to drought.

Human-Induced Causes:

- **Deforestation**: Reduces water retention capacity of soil and disrupts local rainfall patterns.
- **Overexploitation of Water Resources**: Excessive use of groundwater and surface water for irrigation, industries, and domestic purposes.
- **Unsustainable Agriculture**: Over-cultivation and poor irrigation practices deplete water resources.
- **Urbanization**: Expansion of cities reduces natural water infiltration and increases demand for water.

Types of Drought:

1. **Meteorological Drought**:
    1. Occurs when there is a prolonged period of below-average rainfall.
    2. Severity depends on the deviation from normal precipitation.
2. **Hydrological Drought**:
    1. Caused by a significant decrease in water levels in reservoirs, rivers, and groundwater sources.
    2. Impacts water supply for drinking, irrigation, and industries.
3. **Agricultural Drought**:
    1. Results from inadequate soil moisture, affecting crop growth and productivity.
    2. Directly impacts food security.
4. **Socioeconomic Drought**:
    1. Occurs when water shortages affect the economy, leading to reduced income, unemployment, and migration.

Impacts of Drought:

1. **Environmental Impacts**:
    1. Depletion of water bodies and loss of aquatic ecosystems.
    2. Soil degradation, desertification, and loss of biodiversity.
2. **Social Impacts**:
    1. Scarcity of drinking water and food.
    2. Migration and displacement of rural communities.
3. **Economic Impacts**:
    1. Reduced agricultural yield leads to higher food prices.
    2. Loss of livelihood for farmers and workers in water-dependent industries.
4. **Health Impacts**:
    1. Malnutrition due to food shortages.
    2. Increased prevalence of waterborne diseases and poor sanitation.

QUESTION

Explain briefly about Cyclones, its causes, impacts and mitigation.

Volcanoes

Definition:

A volcano is an opening or rupture in the Earth's crust through which molten rock (magma), ash, and gases escape to the surface. Volcanic eruptions can vary in intensity, from gentle lava flows to violent explosions.

Causes of Volcanic Eruptions:

1. **Tectonic Plate Movements**:
    1. Volcanoes are often located at plate boundaries where plates converge (subduction zones) or diverge (mid-ocean ridges).
2. **Magma Formation**:
    1. Magma forms when rock melts due to increased temperature, decreased pressure, or the presence of water in subduction zones.
3. **Pressure Build-up**:
    1. Magma rises to the surface due to its lower density compared to surrounding rock, and pressure buildup in magma chambers causes eruptions.
4. **Hot Spots**:
    1. Volcanic activity can occur away from plate boundaries at mantle hot spots (e.g., Hawaii).

Impacts of Volcanic Eruptions:

1. **Environmental Impacts**:
    1. Destruction of vegetation, wildlife, and ecosystems.
    2. Emission of gases (e.g., CO_2, SO_2) contributes to air pollution and climate change.
    3. Formation of fertile soils over time due to ash deposition.
2. **Social Impacts**:
    1. Loss of life and displacement of communities living near volcanoes.
    2. Disruption of transportation and communication due to ash clouds.
3. **Economic Impacts**:
    1. Damage to infrastructure, homes, and industries.
    2. Tourism can either decline due to safety concerns or increase due to curiosity about the volcano.
4. **Global Impacts**:
    1. Large eruptions can influence global temperatures (e.g., cooling effect from ash and sulfur aerosols blocking sunlight).

Mitigation and Management:

1. **Monitoring and Early Warning Systems**:
    1. Use of seismographs, gas analyzers, and satellite imagery to predict eruptions.
2. **Evacuation Plans**:
    1. Preparedness measures to evacuate people from high-risk areas.
3. **Zoning and Land-Use Planning**:
    1. Restricting settlement near active volcanoes.
4. **Education and Awareness**:
    1. Informing communities about volcanic hazards and safety measures.
5. **Volcanic Eruption Relief**:
    1. Providing emergency aid, rebuilding infrastructure, and offering financial assistance to affected communities.

Landslides

Definition:

Landslides are the downward movement of soil, rock, and debris along a slope, driven by gravity. They are triggered by natural factors like heavy rainfall or earthquakes, as well as human activities such as deforestation and unplanned construction

Causes of Landslides:

1. **Natural Causes**:
    1. **Heavy Rainfall**:
        1. Prolonged or intense rainfall saturates the soil, reducing its stability.
    2. **Earthquakes**:
        1. Ground shaking causes slope instability, leading to landslides.
    3. **Volcanic Activity**:
        1. Eruptions cause debris flows and loosen soil on slopes.
    4. **Snowmelt**:
        1. Rapid melting of snow increases water content in soil, triggering landslides.

1. **Human-Induced Causes**:
    1. **Deforestation**:
        1. Removal of vegetation reduces the binding of soil, making slopes more prone to sliding.
    2. **Unplanned Construction**:
        1. Road cutting and mining activities disturb the slope's natural balance.
    3. **Overloading Slopes**:
        1. Construction or agriculture on steep slopes increases pressure and instability.
    4. **Improper Drainage**:
        1. Poor drainage systems lead to water accumulation, reducing soil strength.

Types of Landslides:

1. **Falls**:
    1. Free-fall of rocks or debris from a steep slope due to gravity.
    2. Example: Rockfalls in hilly regions.
2. **Slides**:
    1. Movement of soil or rock masses along a well-defined surface.
    2. Subtypes:
        1. **Rotational Slides**: Curved movement of material.
        2. **Translational Slides**: Movement along a flat surface.
3. **Flows**:
    1. Rapid movement of soil and debris mixed with water, resembling a liquid flow.
    2. Example: Mudflows and debris flows.
4. **Creeps**:
    1. Slow, gradual movement of soil or rock down a slope.
    2. Often unnoticed until long-term impacts are visible.

Impacts of Landslides:

1. **Environmental Impacts**:
    1. Destruction of vegetation and wildlife habitats.
    2. Soil erosion and alteration of landscapes.
    3. Siltation of rivers and reservoirs.
2. **Social Impacts**:
    1. Loss of life and injuries due to sudden landslides.
    2. Displacement of communities in affected regions.
3. **Economic Impacts**:
    1. Damage to infrastructure like roads, railways, and buildings.
    2. High costs for rehabilitation and reconstruction.
4. **Secondary Impacts**:
    1. Landslides can block rivers, leading to floods.
    2. Triggers other disasters such as avalanches in snowy regions.

Mitigation and Management of Landslides:

1. **Prevention Measures**:
    1. **Afforestation**: Planting trees to stabilize slopes.
    2. **Drainage Control**: Constructing proper drainage systems to prevent water accumulation.
    3. **Retaining Walls**: Building walls to hold back soil and reduce the risk of slides.
    4. **Terracing**: Creating stepped slopes to reduce water flow and erosion.
2. **Monitoring and Early Warning**:
    1. Use of geotechnical instruments to monitor slope stability.
    2. Satellite-based monitoring for landslide-prone areas.
3. **Preparedness**:
    1. Educating communities about landslide risks and safety measures.
    2. Developing evacuation plans for high-risk areas.
4. **Response**:
    1. Rapid deployment of rescue teams to save lives and provide medical aid.
    2. Clearing debris to restore transportation and communication.
5. **Recovery**:
    1. Rebuilding damaged infrastructure.
    2. Relocating affected populations to safer areas.

Coastal Erosion

Definition:

Coastal erosion is the process of the gradual removal of coastal land or beach sediments by natural forces such as waves, tides, currents, and wind. It leads to the loss of landmass along the shoreline and poses a significant threat to coastal ecosystems, human settlements, and infrastructure.

Causes of Coastal Erosion:

1. **Natural Causes**:
    1. **Wave Action**:
        1. Continuous pounding of waves wears away coastal land.
    2. **Tidal Currents**:
        1. Strong tidal movements transport sediments away from the shore.
    3. **Sea-Level Rise**:
        1. Rising sea levels, often linked to climate change, submerge and erode coastal areas.
    4. **Storm Surges**:
        1. Cyclones and storms cause sudden and intense erosion by increasing wave energy.
    5. **Wind Action**:
        1. Wind carries away fine sediments from beaches and dunes.

Human-Induced Causes:

- **Sand Mining**:
  - Excessive extraction of sand disrupts sediment balance, leading to erosion.
- **Construction of Dams**:
  - Dams reduce sediment flow to coastal areas, increasing erosion downstream.
- **Unplanned Coastal Development**:
  - Urbanization and infrastructure development disturb natural coastal processes.
- **Removal of Vegetation**:
  - Loss of mangroves and coastal forests reduces natural protection against erosion.

Impacts of Coastal Erosion:

1. **Environmental Impacts**:
    1. Loss of coastal habitats, including mangroves, coral reefs, and wetlands.
    2. Degradation of ecosystems affecting biodiversity.
    3. Increased salinization of soil, making coastal land unsuitable for agriculture.
2. **Social Impacts**:
    1. Displacement of communities living in coastal areas.
    2. Loss of cultural and historical landmarks along the shore.
3. **Economic Impacts**:
    1. Damage to coastal infrastructure such as roads, ports, and buildings.
    2. Adverse effects on tourism and fisheries.
4. **Increased Vulnerability**:
    1. Coastal areas become more exposed to flooding and storm surges due to loss of natural barriers.

Mitigation and Management of Coastal Erosion:

1. **Preventive Measures**:
    1. **Afforestation**:
        1. Planting mangroves and other vegetation to stabilize coastal land.
    2. **Beach Nourishment**:
        1. Adding sand or sediments to eroded beaches to restore their shape.
    3. **Construction of Breakwaters and Groynes**:
        1. Structures built to reduce wave energy and prevent sediment movement.
2. **Sustainable Development**:
    1. Regulating construction activities and promoting eco-friendly infrastructure in coastal zones
    2. Prohibiting activities like sand mining in sensitive areas.
3. **Climate Change Mitigation**:
    1. Reducing greenhouse gas emissions to slow down sea-level rise.
    2. Implementing policies to adapt to the impacts of climate change.

4. Community Awareness:

1. Educating local populations about the causes and impacts of coastal erosion.
2. Encouraging participation in conservation programs.

5. Monitoring and Research:

1. Using satellite imagery and GIS tools to track erosion patterns.
2. Conducting studies to understand the effectiveness of mitigation measures.

Soil Erosion

Definition:

Soil erosion is the process by which the top layer of soil is removed or displaced by natural forces such as water, wind, and ice, or by human activities like deforestation and agriculture. It results in the loss of fertile land and can severely impact agriculture, ecosystems, and water quality.

Causes of Soil Erosion:

1. **Natural Causes**:
    1. **Water Erosion**:
        1. Rainfall and surface runoff wash away the topsoil.
        2. Types include sheet erosion, rill erosion, and gully erosion.
    2. **Wind Erosion**:
        1. Strong winds blow away loose topsoil in arid and semi-arid regions.
    3. **Glacial Erosion**:
        1. Movement of glaciers scrapes soil and rocks from the surface.
    4. **Gravity**:
        1. Landslides and slope movement transport soil downslope.

1. **Human-Induced Causes**:
    1. **Deforestation**:
        1. Removal of trees reduces soil stability and increases vulnerability to erosion.
    2. **Overgrazing**:
        1. Excessive grazing by livestock exposes soil, making it prone to erosion.
    3. **Unsustainable Agriculture**:
        1. Practices like mono-cropping, improper plowing, and over-cultivation degrade soil.
    4. **Urbanization**:
        1. Construction activities disturb natural soil cover and increase runoff.
    5. **Mining Activities**:
        1. Strip mining and quarrying lead to large-scale soil displacement.

Types of Soil Erosion:

1. **Sheet Erosion**:
    1. Removal of a thin, uniform layer of soil across a large area due to surface runoff.
2. **Rill Erosion**:
    1. Formation of small channels on the soil surface by running water.
3. **Gully Erosion**:
    1. Large channels or gullies form as water removes significant amounts of soil.
4. **Wind Erosion**:
    1. Loose, dry soil particles are carried away by the wind, common in deserts and semi-arid regions.
5. **Streambank Erosion**:
    1. Erosion of soil along riverbanks due to flowing water.

Impacts of Soil Erosion:

1. **Environmental Impacts**:
    1. **Loss of Fertile Topsoil**:
        1. Reduces agricultural productivity and soil fertility.
    2. **Sedimentation of Water Bodies**:
        1. Eroded soil is deposited in rivers, lakes, and reservoirs, affecting aquatic ecosystems.
    3. **Desertification**:
        1. Prolonged soil erosion in arid areas leads to the transformation of fertile land into deserts.
2. **Economic Impacts**:
    1. Decline in crop yields and higher costs for fertilizers and soil restoration.
    2. Loss of arable land, impacting livelihoods dependent on agriculture.
3. **Social Impacts**:
    1. Migration of communities due to unproductive land.
    2. Increased competition for fertile land and water resources.
4. **Climate Impacts**:
    1. Erosion contributes to the release of carbon stored in soil, exacerbating climate change.

Mitigation and Management of Soil Erosion:

1. **Preventive Measures**:
    1. **Afforestation and Reforestation**:
        1. Planting trees and vegetation to stabilize soil and reduce runoff.
    2. **Terracing**:
        1. Creating step-like structures on slopes to reduce water flow and erosion.
    3. **Contour Plowing**:
        1. Plowing along natural contours to reduce soil loss.
    4. **Mulching**:
        1. Using organic materials like straw or leaves to cover soil and prevent erosion.
2. **Sustainable Farming Practices**:
    1. **Crop Rotation**:
        1. Alternating crops to maintain soil fertility.
    2. **Cover Cropping**:
        1. Growing cover crops to protect soil during the off-season.

3. **Engineering Solutions**:
    1. **Check Dams**:
        1. Small dams built across gullies to slow water flow and reduce erosion.
    2. **Retaining Walls**:
        1. Structures that prevent soil movement on steep slopes.
4. **Community and Policy Measures**:
    1. **Soil Conservation Programs**:
        1. Government initiatives to promote soil health and conservation.
    2. **Awareness Campaigns**:
        1. Educating farmers and communities on soil erosion and prevention methods.

Forest Fires

Definition:

Forest fires, also known as wildfires, are uncontrolled fires that spread rapidly through vegetation, forests, and grasslands. These fires can cause significant environmental, social, and economic damage.

Causes of Forest Fires:

1. **Natural Causes**:
    1. **Lightning**:
        1. A common natural trigger for forest fires, especially in dry regions.
    2. **Volcanic Eruptions**:
        1. Lava flows and ash can ignite surrounding vegetation.
    3. **Spontaneous Combustion**:
        1. Dry vegetation and organic matter can ignite under high temperatures and pressure.
2. **Human-Induced Causes**:
    1. **Deforestation**:
        1. Slash-and-burn practices often lead to unintended forest fires.
    2. **Agricultural Activities**:
        1. Clearing land for farming using fire can result in uncontrolled blazes.
    3. **Negligence**:
        1. Campfires, discarded cigarettes, and unattended burning debris.
    4. **Arson**:
        1. Deliberate setting of fires for illegal purposes like land encroachment.

Types of Forest Fires:

1. **Surface Fires**:
    1. Burn through the undergrowth, grass, and fallen leaves without affecting trees significantly.
    2. Easier to control compared to other types.
2. **Crown Fires**:
    1. Spread through the canopy or treetops.
    2. Often caused by strong winds and are highly destructive.
3. **Ground Fires**:
    1. Burn beneath the surface, affecting peat and organic matter.
    2. Difficult to detect and extinguish.

Impacts of Forest Fires:

1. **Environmental Impacts**:
    1. **Loss of Vegetation**:
        1. Destroys forests, reducing biodiversity and wildlife habitats.
    2. **Air Pollution**:
        1. Smoke and gases (e.g., carbon dioxide, methane) contribute to air pollution and climate change.
    3. **Soil Degradation**:
        1. High-intensity fires burn organic matter, reducing soil fertility.
    4. **Water Contamination**:
        1. Ash and debris can contaminate nearby water bodies.
2. **Social Impacts**:
    1. **Human Displacement**:
        1. People living in affected areas may lose homes and livelihoods.
    2. **Health Issues**:
        1. Smoke inhalation causes respiratory problems and other health risks.
3. **Economic Impacts**:
    1. **Loss of Timber and Resources**:
        1. Forest resources are destroyed, affecting industries like logging.
    2. **High Costs of Firefighting**:
        1. Governments spend large amounts on firefighting operations and rehabilitation.
4. **Global Impacts**:
    1. **Climate Change**:
        1. Releases large amounts of carbon dioxide, contributing to global warming.

Mitigation and Management of Forest Fires:

1. **Prevention Measures**:
    1. **Controlled Burning**:
        1. Planned fires to reduce dry vegetation and prevent larger wildfires.
    2. **Firebreaks**:
        1. Clearing strips of land to stop fire spread.
    3. **Awareness Campaigns**:
        1. Educating communities on fire safety and prevention.
2. **Monitoring and Early Warning**:
    1. **Satellite Systems**:
        1. Using satellites to monitor and detect forest fires in real-time.
    2. **Fire Weather Forecasting**:
        1. Predicting conditions that increase fire risk, like high temperatures and wind.
3. **Preparedness**:
    1. **Community Training**:
        1. Training local communities and firefighters in fire management techniques.
    2. **Evacuation Plans**:
        1. Establishing clear evacuation protocols for affected areas.

5. **Response**:
    1. **Firefighting Operations**:
        1. Deploying fire suppression teams, aircraft, and equipment to control the fire.
    2. **Emergency Aid**:
        1. Providing shelter, food, and medical assistance to affected populations.
6. **Recovery**:
    1. **Reforestation**:
        1. Planting trees and restoring damaged ecosystems.
    2. **Rehabilitation of Communities**:
        1. Rebuilding infrastructure and supporting livelihoods.

Man-Made Disasters

Definition:

Man-made disasters are catastrophic events caused by human activities, leading to loss of life, environmental damage, and economic losses. These disasters can result from negligence, accidents, or intentional actions.

Types of Man-Made Disasters:

1. Industrial Accidents

Definition:

Industrial accidents refer to unexpected events occurring in factories, power plants, or industrial facilities that result in injury, loss of life, environmental damage, and economic losses. These accidents may be caused by human error, technical failures, hazardous material leaks, or inadequate safety measures.

Causes of Industrial Accidents:

1.Human Error:

- Poor training, lack of awareness, or negligence in handling machinery and hazardous substances.
- Mismanagement of industrial processes leading to operational failures.

2. Mechanical Failures:

•Malfunctioning equipment, outdated machinery, or poor maintenance can lead to dangerous failures.

•Examples include boiler explosions, machinery breakdowns, or pipeline leaks.

3. Chemical and Gas Leaks:

•Accidental release of toxic gases or chemicals due to improper storage, handling, or equipment failure.

•Can cause severe health hazards, contamination, and long-term environmental damage.

4. Fires and Explosions:

•Flammable materials, electrical faults, and gas leaks can result in industrial fires and explosions.

•Lack of fire safety measures worsens the impact.

5. Poor Safety Standards and Regulation Violations:

•Inadequate adherence to industrial safety protocols.

•Insufficient protective gear, lack of emergency exits, and absence of firefighting equipment.

6. Environmental Factors:

•Natural disasters such as earthquakes, floods, and storms can trigger industrial accidents.

•Example: A tsunami led to the Fukushima nuclear disaster in Japan (2011).

Types of Industrial Accidents:

Chemical Accidents:

Involves the release of toxic chemicals, causing air, water, and soil pollution.

Example: Bhopal Gas Tragedy (1984) due to methyl isocyanate gas leakage.

Nuclear Accidents:

Radiation leaks from nuclear power plants can have long-term effects on health and the environment.

Example: Chernobyl Disaster (1986).

Machinery-Related Accidents:

Workers getting injured due to malfunctioning or improperly used machines.

Example: Factory workers getting caught in conveyor belts or presses.

Fires and Explosions:

Often caused by gas leaks, overheating equipment, or electrical faults.

Example: Texas City Refinery explosion (2005).

Impacts of Industrial Accidents:

Human Impact:

Loss of lives, injuries, and long-term health issues such as respiratory diseases and cancer due to toxic exposure.

Psychological trauma among affected workers and communities.

Environmental Impact:

Contamination of air, water, and soil.

Long-term ecological damage, leading to biodiversity loss and habitat destruction.

Economic Impact:

Loss of industrial production and huge financial losses for companies.

Compensation claims, legal actions, and regulatory fines.

Social Impact:

Displacement of communities due to hazardous contamination.

Distrust in industrial development leading to protests and opposition to projects.

Mitigation and Prevention of Industrial Accidents:

Strict Safety Regulations:
Enforcing workplace safety laws and compliance with environmental standards.
Regular inspection of industrial facilities.

Worker Training and Awareness:
Conducting safety drills, training workers on emergency protocols, and using personal protective equipment (PPE).

Use of Advanced Technology:
Installing safety alarms, automated shut-off systems, and sensors to detect gas leaks or overheating.

Emergency Response Plans:
Creating disaster management teams and equipping factories with fire extinguishers, emergency exits, and medical aid stations.

Sustainable Industrial Practices:
Reducing hazardous waste, using environmentally friendly materials, and adopting cleaner technologies.

2. Pollution-Related Disasters

Definition: Environmental damage caused by excessive release of pollutants into air, water, or land.

Types:

- **Air Pollution** (e.g., smog in urban areas).
- **Water Pollution** (e.g., oil spills, industrial waste dumping).
- **Soil Pollution** (e.g., pesticide overuse, hazardous waste disposal).

Example: Deepwater Horizon oil spill (2010) caused severe marine pollution.

3. Urban Fires

Definition: Fires occurring in densely populated areas due to electrical faults, gas leaks, or flammable material storage.

Examples:

- **Grenfell Tower Fire (2017, UK)**: A high-rise apartment fire caused by flammable cladding.
- **Kamala Mills Fire (2017, Mumbai, India)**: A fire in a commercial building due to a short circuit.

4. Transportation Accidents

Definition: Disasters related to road, rail, air, or maritime transportation due to human error, mechanical failure, or natural hazards.

Types:

- **Road Accidents** (e.g., vehicle collisions due to reckless driving).
- **Rail Accidents** (e.g., train derailments, track failures).
- **Airplane Crashes** (e.g., pilot error, bad weather, mechanical failures).
- **Shipwrecks and Oil Spills** (e.g., Titanic sinking in 1912, MV Wakashio oil spill in 2020).

5. Terrorism and Bomb Blasts

Definition: Acts of violence committed by extremist groups to create fear and instability.

Examples:

- **9/11 Attacks (2001, USA)**: Terrorist attack on the World Trade Center.
- **26/11 Mumbai Attacks (2008, India)**: Coordinated terrorist attacks in multiple locations.

6. Nuclear and Radiological Disasters

Definition: Accidents involving the release of harmful radioactive materials, often due to nuclear power plant failures or improper waste disposal.

Examples:

- **Chernobyl Disaster (1986)**: Explosion at a nuclear power plant caused radiation exposure.
- **Fukushima Nuclear Disaster (2011, Japan)**: Tsunami triggered nuclear meltdown.

7. Cyber Attacks

Definition: Unauthorized digital intrusions that compromise security, steal information, or disrupt critical services.

Examples:

- **WannaCry Ransomware Attack (2017)**: Cyberattack affecting thousands of systems worldwide.
- **Data Breaches in Banking and Healthcare Sectors**.

Hazard and Vulnerability Profile of India

Introduction

India is one of the most disaster-prone countries in the world due to its diverse geographical and climatic conditions. It is vulnerable to various natural and man-made hazards, including earthquakes, floods, droughts, cyclones, landslides, and industrial accidents. Understanding the hazard and vulnerability profile of India helps in better disaster management and risk reduction strategies.

Types of Hazards in India

India is prone to multiple hazards due to its location, climate, and topographical features.

(a) Earthquakes

•India lies in a seismically active region, with many areas falling under **seismic zones IV and V**, which are highly earthquake-prone.

•**Major Earthquake-Prone Areas:**

- Himalayan region (Jammu & Kashmir, Himachal Pradesh, Uttarakhand, Arunachal Pradesh)
- Northeastern states (Assam, Nagaland, Manipur, Mizoram, Tripura)
- Indo-Gangetic Plain (Delhi, Bihar, Uttar Pradesh)
- Western India (Gujarat, Maharashtra)

•**Major Earthquakes in India:Bhuj Earthquake (2001)** – Gujarat, 7.7 magnitude, caused over 20,000 deaths.

•**Kashmir Earthquake (2005)** – Affected India and Pakistan, over 80,000 deaths.

(b) Floods

India experiences **frequent and severe floods** due to heavy monsoon rains and river overflow.

Flood-Prone Areas:

- Ganga and Brahmaputra river basins (Uttar Pradesh, Bihar, Assam, West Bengal)
- Coastal states (Odisha, Andhra Pradesh, Tamil Nadu, Kerala)
- Cities with poor drainage (Mumbai, Chennai, Hyderabad, Bengaluru)

Causes of Floods:

- Heavy monsoon rains
- Poor drainage and encroachments on floodplains
- Deforestation and soil erosion

Recent Major Floods:

- **Assam Floods (2020, 2022)** – Displaced millions of people.
- **Kerala Floods (2018, 2019)** – Caused by excessive rainfall and dam releases.

(c) Droughts

•Droughts occur due to **low rainfall and water scarcity**, affecting agriculture and drinking water supply.

•**Drought-Prone Areas:**

- Rajasthan, Gujarat, Maharashtra, Karnataka, Andhra Pradesh, Tamil Nadu.

•**Causes of Droughts:**

- Weak monsoon
- Overuse of groundwater
- Climate change and deforestation

•**Impact of Droughts:**

- Crop failure, food shortages, farmer suicides.
- Example: **Marathwada Drought (2016, Maharashtra) –** Affected millions of farmers.

(d) Cyclones

•Coastal states of India are highly **prone to cyclones** from the **Bay of Bengal and the Arabian Sea**.

•**Cyclone-Prone States:**

- Odisha, West Bengal, Andhra Pradesh, Tamil Nadu, Gujarat, Maharashtra.

•**Major Cyclones in India:**

- **Cyclone Amphan (2020) –** Affected West Bengal and Odisha, caused massive destruction.
- **Cyclone Tauktae (2021) –** Hit Gujarat and Maharashtra, leading to severe flooding and loss of life.

(e) Landslides

Common in **hilly regions** due to heavy rainfall, earthquakes, and deforestation.

Landslide-Prone Areas:

- Himalayan region (Himachal Pradesh, Uttarakhand, Sikkim, Arunachal Pradesh)
- Western Ghats (Kerala, Maharashtra)

Causes of Landslides:

- Deforestation
- Unplanned construction
- Heavy rainfall and earthquakes

Recent Major Landslides:

- **Himachal Pradesh Landslides (2023)** – Several villages buried due to continuous heavy rains.
- **Malin Landslide (2014, Maharashtra)** – Over 150 people buried under debris.

(f) Tsunamis

Coastal areas of **Tamil Nadu, Andhra Pradesh, Odisha, Andaman & Nicobar Islands** are vulnerable to tsunamis.

Cause: Underwater earthquakes and volcanic eruptions.

Example: **2004 Indian Ocean Tsunami** – Affected Tamil Nadu, Andaman & Nicobar, killing over 10,000 people in India.

Vulnerability Profile of India

India's vulnerability to disasters is due to several **physical, social, and economic** factors.

1. Physical Vulnerability

Geographical Location:

Lies on active seismic zones and monsoon-affected regions.

Long coastline (over 7,500 km) makes it highly susceptible to cyclones, tsunamis, and erosion.

2. Social Vulnerability

High Population Density:

Over 1.4 billion people, with many living in disaster-prone areas.

Poverty and Lack of Infrastructure:

Poor housing, sanitation, and healthcare systems increase vulnerability.

Urbanization:

Unplanned growth leads to waterlogging, pollution, and slum formation.

3. Economic Vulnerability

Dependence on Agriculture:

Over 60% of the population relies on agriculture, which is highly affected by droughts, floods, and climate change.

Poor Disaster Preparedness:

Limited resources for disaster risk reduction in rural areas.

Government Initiatives for Disaster Management

The Government of India has taken several steps to **reduce disaster risks and improve preparedness**.

1. **National Disaster Management Authority (NDMA)**:
   Established in 2005 to formulate policies and coordinate disaster response.
2. **Disaster Early Warning Systems**:
   Indian Meteorological Department (IMD) provides cyclone, flood, and earthquake warnings.
3. **State Disaster Response Forces (SDRF) & National Disaster Response Force (NDRF)**:
   Specialized teams for disaster rescue and relief.
4. **Coastal Regulation Zone (CRZ) Rules**:
   Restricts construction and industrial activities along vulnerable coastal areas.
5. **Smart Cities Mission**:
   Focuses on climate-resilient infrastructure to reduce disaster risks.

Caved 2019 in India

The term **"caved"** refers to a **collapse or sinking of the ground, structures, or underground spaces** due to natural or man-made causes.

In 2019, India experienced several incidents involving cave-ins and related disasters. Notable events include:

1. Meghalaya Mining Disaster (December 2018 – January 2019):

•**Location:** Ksan village, East Jaintia Hills district, Meghalaya.

•**Incident:** On December 13, 2018, 15 miners were trapped in a 370-foot-deep illegal "rat-hole" coal mine when water from the nearby Lytein River flooded the shafts.

•**Challenges:** Rescue operations were hampered by inadequate equipment, lack of a detailed mine map, and the depth of flooding.

•**Outcome:** Despite efforts, only two bodies were recovered; the remaining miners were declared deceased after weeks of unsuccessful rescue attempts.

2. Guna Caves Incidents:
•**Location:** Kodaikanal, Tamil Nadu.
•**Background:** Known as Devil's Kitchen, Guna Caves gained popularity after the 1991 Tamil film "Gunaa."
•**Incidents:** The caves have a history of accidents due to their narrow and deep passages. As of 2016, police records indicated 16 disappearances linked to the caves, with bodies unrecovered.
•**Safety Measures:** Authorities have restricted access to the caves to prevent further accidents.

3. Gurugram-Sohna Road Cave-In (July 2019):
•**Location:** Gurugram, Haryana.
•**Incident:** A section of the Gurugram-Sohna elevated flyover caved in, creating a significant pothole.
•**Impact:** A school bus narrowly avoided a major accident when one of its wheels got stuck in the caved-in portion.
•**Cause:** Preliminary reports suggested the cave-in resulted from issues related to a sewer line meant for road drainage.

Ecological Fragility

Definition:

Ecological fragility refers to the vulnerability of ecosystems to disturbances caused by **natural factors** (such as climate change, natural disasters) and **human activities** (such as deforestation, pollution, and urbanization). Fragile ecosystems are highly sensitive to changes and take longer to recover from disturbances, leading to biodiversity loss, habitat destruction, and environmental degradation.

Characteristics of Ecologically Fragile Areas:

- **Low resilience** – These ecosystems struggle to recover after disturbances.
- **High sensitivity** – Small environmental changes can lead to significant impacts.
- **Rich biodiversity** – Often home to unique and rare species that are highly dependent on specific environmental conditions.
- **Threatened by human activities** – Unsustainable development, deforestation, and industrial activities can lead to irreversible damage.

Types of Ecologically Fragile Areas in India:

1. Himalayan Ecosystem

•**Threats:**

- Landslides, earthquakes, and glacial retreat due to climate change.
- Deforestation and overgrazing causing soil erosion.
- Hydropower projects disturbing river ecosystems.

•**Example:** Melting of **Gangotri Glacier** threatening the Ganga River's water flow.

2. Western Ghats

•**Threats:**

- Deforestation for agriculture, mining, and urbanization.
- Climate change affecting monsoon patterns and biodiversity.
- Unregulated tourism increasing waste and habitat destruction.

•**Example:** Decline in endemic species like the **Lion-tailed Macaque** due to habitat loss.

3. Coastal and Marine Ecosystems

•**Threats:**

- Rising sea levels and coastal erosion due to climate change.
- Mangrove destruction for aquaculture, tourism, and urban expansion.
- Oil spills and plastic pollution harming marine biodiversity.

•**Example: Sundarbans Mangroves** are disappearing due to sea-level rise and cyclones.

4. Desert Ecosystem (Thar Desert, Rajasthan)

•**Threats:**

- Desertification due to overgrazing and unsustainable agriculture.
- Climate change causing erratic rainfall and loss of water resources.
- Encroachment and mining activities affecting fragile desert flora and fauna.

•**Example: Khejri trees**, vital for desert ecology, are declining due to excessive groundwater use.

5. Forest Ecosystems (Central India, North-East India, Western India)

•**Threats:**

- Illegal logging and deforestation for agriculture and urban expansion.
- Encroachment of industries leading to habitat destruction.
- Poaching and illegal wildlife trade reducing biodiversity.

•**Example: Kaziranga National Park (Assam)** threatened by deforestation and floods.

6. Wetlands and River Ecosystems

•**Threats:**

- Pollution from industrial and domestic waste.
- Shrinking wetlands due to encroachments and urban expansion.
- Dam construction disrupting natural water flow and fish migration.

•**Example: Loktak Lake (Manipur)** is endangered due to human encroachment and pollution.

Causes of Ecological Fragility

1. **Climate Change** – Rising temperatures, changing rainfall patterns, and extreme weather events.
2. **Deforestation** – Large-scale tree cutting for agriculture, urbanization, and infrastructure development.
3. **Industrial Pollution** – Air, water, and soil contamination affecting biodiversity and ecosystem health.
4. **Unregulated Tourism** – Increased human activity leading to habitat destruction and waste generation.
5. **Overexploitation of Natural Resources** – Overfishing, excessive groundwater extraction, and mining.

Impact of Ecological Fragility

1. **Biodiversity Loss –** Extinction of species due to habitat destruction.
2. **Desertification –** Loss of fertile land in semi-arid regions.
3. **Water Scarcity –** Reduced water availability in rivers and groundwater depletion.
4. **Increased Disaster Risk –** Landslides, floods, and forest fires become more frequent.
5. **Climate Change Acceleration –** Carbon-absorbing ecosystems like forests and wetlands are lost, increasing global warming.

Coping with Disaster - Strategies

Coping with disasters involves **preparedness, response, recovery, and mitigation** strategies to minimize damage and ensure quick recovery. Disasters can be **natural (earthquakes, floods, cyclones, etc.)** or **man-made (industrial accidents, fires, pollution, etc.)**, and effective coping mechanisms help reduce their impact on lives, infrastructure, and the environment.

Strategies for Coping with Disasters

1. Prevention and Mitigation Strategies

•Aim to **reduce the likelihood of disasters** and **minimize their impact** when they occur.

•**Examples:**

- **Earthquake-Resistant Buildings** – Using advanced construction techniques in seismic zones.
- **Flood Control Measures** – Building dams, embankments, and improving drainage systems.
- **Afforestation** – Preventing landslides and desertification by planting trees.
- **Industrial Safety Regulations** – Enforcing strict protocols to prevent chemical leaks and explosions.

2. Preparedness Strategies

•Ensuring communities, governments, and organizations are **ready to respond effectively** before a disaster strikes.

•**Key Preparedness Measures:**

- **Early Warning Systems –** Satellite and sensor-based alerts for cyclones, floods, and earthquakes.
- **Disaster Education and Awareness –** Training people on evacuation procedures and first aid.
- **Community-Based Disaster Management (CBDM) –** Local participation in risk assessment and planning.
- **Stockpiling Emergency Supplies –** Food, water, medicines, and safety kits.
- **Mock Drills and Training –** Regular practice for response teams, schools, and workplaces.

3. Response Strategies

•Focuses on **immediate actions** to save lives and provide emergency assistance after a disaster.

•**Key Response Activities:**

- **Search and Rescue Operations –** Deploying National Disaster Response Force (NDRF) teams.
- **Emergency Medical Services –** Providing first aid, hospitals on alert, and mobile clinics.
- **Evacuation and Shelter Management –** Moving affected people to safer locations.
- **Restoration of Essential Services –** Restoring electricity, water supply, and communication networks.
- **Providing Food and Relief Materials –** Distributing ration kits and essential supplies.

4. Recovery Strategies

•Long-term measures taken **after the disaster to restore normalcy** and rebuild affected communities.

•**Key Recovery Steps:**

- **Reconstruction of Infrastructure –** Roads, bridges, hospitals, schools, and homes.
- **Livelihood Rehabilitation –** Providing employment opportunities and financial aid to affected people.
- **Psychosocial Support –** Counseling services for trauma-affected survivors.
- **Insurance and Compensation –** Government compensation schemes and disaster relief funds.

5. Risk Reduction and Long-Term Planning

•**Adopting sustainable policies** to prevent future disasters and minimize risks.

•**Key Risk Reduction Measures:**

- **Zoning Laws and Land-Use Planning –** Avoiding construction in flood-prone or earthquake-sensitive areas.
- **Climate Change Adaptation –** Using renewable energy and reducing carbon emissions.
- **Strengthening Governance –** Implementing better disaster response policies and ensuring funding.

Safety Norms and Survival Kits

1. Safety Norms in Disaster Management

Safety norms are **precautionary measures and guidelines** designed to minimize risks, injuries, and fatalities during disasters. These norms vary based on the type of disaster but share common principles of **preparedness, response, and recovery**.

A. General Safety Norms for Different Disasters

(i) Earthquake Safety Norms

- Construct **earthquake-resistant buildings** using flexible structures.
- Follow the **Drop, Cover, and Hold** technique during tremors.
- Avoid **standing near windows, walls, or furniture** that can fall.
- Have **emergency exits** and conduct regular **earthquake drills**.

(ii) Flood Safety Norms

- **Elevate electrical wiring** and appliances in flood-prone areas.
- Build **embankments and drainage systems** to prevent waterlogging.
- Move to **higher ground** during heavy rainfall warnings.
- Avoid **walking or driving in flooded areas** to prevent drowning.

(iii) Cyclone Safety Norms

•Secure **roofs, doors, and windows** to withstand high winds.
•Evacuate to **cyclone shelters** if issued a warning.
•Keep **emergency communication devices** (radios, mobile phones).

(iv) Fire Safety Norms

•Install **smoke detectors** and **fire extinguishers** in buildings.
•Conduct **fire drills** regularly.
•**Do not use lifts** during a fire; use stairs instead.
•Know the **"Stop, Drop, and Roll"** method if clothes catch fire.

(v) Landslide Safety Norms

•**Avoid construction** on steep slopes.
•Plant **deep-rooted vegetation** to prevent soil erosion.
•Stay away from **unstable cliffs or hill slopes** during heavy rain.

(vi) Industrial and Chemical Disaster Safety Norms

- **Proper storage** of hazardous chemicals to prevent leaks.
- Workers must wear **Personal Protective Equipment (PPE)**.
- Install **gas leak detectors and ventilation systems**.
- Evacuate immediately in case of gas leaks and seek medical help.

(vii) Road and Transport Safety Norms

- Follow **traffic rules** and wear **seat belts/helmets**.
- Regular **vehicle maintenance** to avoid mechanical failures.
- Strict **speed limits and DUI (Driving Under Influence) laws**.

Survival Kits for Disaster Preparedness

A **survival kit** is an essential collection of emergency supplies designed to help individuals and families sustain themselves during and after a disaster. It should be **compact, portable, and contain basic survival essentials**.

A. Essential Items in a Survival Kit

(i) Basic Necessities

- **Drinking water** – At least 3 liters per person per day.
- **Non-perishable food** – Canned food, energy bars, dry fruits.

(ii) First Aid and Medical Supplies

- **First aid kit** – Bandages, antiseptics, pain relievers, scissors.
- **Essential medicines** – For chronic illnesses (diabetes, BP, asthma).
- **Hand sanitizer and disinfectants** – To prevent infections.

(iii) Emergency Tools and Equipment

- **Battery-powered torch/flashlight** – With extra batteries.
- **Whistle** – To signal rescuers in case of entrapment.
- **Multi-purpose knife or tool kit** – For minor repairs.
- **Firestarter/matches/lighter** – For cooking or warmth.

(iv) Communication and Navigation

- **Mobile phone with power bank –** For emergency communication.
- **Battery-powered or solar radio –** To receive weather alerts.
- **Map and compass –** For navigation in case of displacement.

(v) Shelter and Clothing

- **Blankets and sleeping bags –** To protect against cold.
- **Waterproof clothing and raincoats –** For protection against harsh weather.
- **Sturdy shoes –** Essential for rough terrain or walking long distances.

(vi) Important Documents

- **Identity proof, insurance papers, bank documents** (stored in waterproof pouches).
- **Emergency contact numbers and addresses**.

(vii) Personal Hygiene Items

- **Soap, toothpaste, toothbrush, and sanitary items**.
- **Tissues and wet wipes** for cleanliness.

(viii) Special Items (Based on Individual Needs)

- **Baby food and diapers –** If infants are part of the family.
- **Pet food and leash –** If traveling with pets.